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DEPARTMENT OF CSE(DATA SCIENCE)

A Mini-Project Report On

"Netflix Data Analysis Dashboard using Power BI "

A report submitted in partial fulfillment of the requirements for the

DATA VISUALIZATION WITH POWER BI

Submitted By

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Belagavi, Karnataka 2025-2026

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CERTIFICATE

This is to certify that the OPEN-ENDED PROJECT of DATA VISUALIZATION WITH POWER BI **"Netflix Data Analysis Dashboard using Power BI"** has been successfully presented by **G.L SRI NIDHI** bearing USN **3BR23CD021** student of semester B.E for the partial fulfillment of the requirements for the award of **Bachelor Degree in CSE(DS)** of the BALLARI INSTITUTE OF TECHNOLOGY& MANAGEMENT, BALLARI during the academic year 2025-2026.

Signature of guide	Signature of HOD
Mr. Sundeep Sindhanoor	**Dr. Aradhana D**

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ABSTRACT

The “Netflix Data Analysis Dashboard using Power BI” project is developed to provide a comprehensive, interactive, and visually appealing analysis of Netflix’s movies and TV shows dataset. With the rapid growth of digital streaming platforms, Netflix has accumulated a vast amount of content across different genres, countries, and categories, making it difficult to analyze trends using traditional methods. This project addresses that challenge by transforming raw Netflix data into meaningful insights through data visualization techniques.The dataset used in this project consists of information such as title, type (Movie or TV Show), genre, country, rating, and release year. Using Power BI, the data is cleaned, transformed, and modeled to ensure accuracy and consistency. Power Query is utilized for data preprocessing tasks such as handling missing values, formatting columns, and structuring the dataset, while DAX (Data Analysis Expressions) is used to create calculated measures like total titles, genre count, and rating distribution.The dashboard includes various visualizations such as KPI cards, bar charts, donut charts, treemaps, and line graphs to represent different aspects of the data. KPI cards provide a quick overview of key metrics like total titles, total genres, and total ratings. Bar charts are used to display genre distribution, treemaps represent country-wise content contribution, and donut charts compare the proportion of movies and TV shows. Line charts are used to analyze trends in content growth over the years.

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CHAPTER 1

INTRODUCTION

1.1 About the Project

In recent years, the rapid growth of digital streaming platforms has transformed the way people consume entertainment. Among these platforms, Netflix has emerged as one of the leading global providers of movies and TV shows, offering a vast library of content across multiple genres, languages, and regions. With such a large and continuously expanding dataset, it becomes challenging to analyze and extract meaningful insights using traditional methods. This creates a need for efficient data analysis and visualization tools that can simplify complex data and present it in an understandable format.

The "Netflix Data Analysis Dashboard using Power BI" project focuses on analyzing Netflix content data and presenting key insights through an interactive dashboard. Power BI, a powerful business intelligence and data visualization tool developed by Microsoft, is used to transform raw data into visually rich and interactive reports. This project aims to explore various aspects of Netflix data, such as content distribution by genre, country-wise production, ratings classification, and trends in content growth over the years.

Overall, this project highlights the importance of data visualization in modern data analysis. By converting raw data into interactive visual insights, it enables better understanding, faster decision-making, and effective communication of information. The Netflix dashboard serves as a practical example of how business intelligence tools like Power BI can be used to analyze real-world datasets and uncover valuable insights.

1.2 Language Used

The core tools used in this project are:

- **Power BI:** Used for data modeling, formatting, dashboard creation and visualisation.
- **DAX (Data Analysis Expressions):** Used in Power BI to create dynamic calculations, measures, and KPIs such as YoY changes, average carbon intensity, and total emissions.
- **CSV (Data Source Format):** The primary data source format used to import the global energy dataset into Power BI for further processing and visualization

Together, these tools form a cohesive system that efficiently handles data transformation and analysis, ultimately delivering clear, actionable insights through a user-friendly dashboard.

CHAPTER 2

SCOPE OF THE PROJECT

2.1 Problem Statement

With the rapid growth of Netflix as a global streaming platform, a massive amount of data related to movies and TV shows is generated, including details about genres, countries, ratings, and release years. However, analyzing this large and complex dataset using traditional methods is time-consuming and inefficient. It becomes difficult to identify patterns, trends, and key insights such as popular genres, dominant countries, and content growth over time without proper visualization tools.

Therefore, there is a need for an efficient system that can convert raw Netflix data into meaningful and easily understandable insights. The problem is addressed by developing an interactive Power BI dashboard that visually represents the data through charts and graphs. This enables users to quickly analyze trends, compare categories, and make data-driven decisions in a simple and effective manner.

2.2 Objectives

The main objectives of the project are as follows:

- Analyze Netflix dataset
- Visualize content distribution
- Identify popular genres and ratings
- Study content growth over years
- Provide interactive dashboard

CHAPTER 3
SYSTEM REQUIREMENTS

3.1 Functional Requirements

- Import Netflix dataset from CSV/Excel into Power BI for analysis.
- Perform data cleaning and transformation using Power Query to ensure accurate and structured data.
- Create DAX measures to calculate key metrics such as total titles, genres, and ratings.
- Develop interactive visualizations like bar charts, treemaps, donut charts.
- Implement filters and slicers to allow users to dynamically explore data based on year, country.

3.2 Non-Functional Requirements

- **Performance:** Dashboard interactions, filters, and visuals should load and respond quickly (within ~3–5 seconds) for a smooth user experience.
- **Usability:** The interface should be intuitive, clean, and easy to navigate, with clear labels, readable visuals, and proper color contrast.
- **Scalability:** The data model should support future data additions (e.g., beyond 2024) without significant performance issues.
- **Maintainability:** DAX measures and Power Query (M) steps should be well-structured
- **Security:** Access to the dashboard and dataset should be restricted to authorized users, especially when deployed on Power BI Service.
- **Reliability:** The dashboard should consistently provide accurate results and stable performance without errors or failures.

3.3 Hardware Requirements

- **Operating System:** Windows 10, Windows 11, or Windows Server 2016/2019/2022.
- **Processor (CPU):** 1.4 GHz 64-bit (x64) processor or better.
- **Memory (RAM):** 4 GB RAM.
- **Storage:** 2 GB of available hard disk space (for the Power BI Desktop application and cache).
- **Display:** 1440 x 900 or 1600 x 900 (16:9) resolution.

3.4 Software Requirements

- **Development:** Microsoft Power BI Desktop (requires Windows 10/11).
- **Data Source:** CSV files (can be managed via Microsoft Excel or standard text editors).
- **Deployment & Hosting:** Power BI Service (cloud platform for publishing and sharing the dashboard).
- **End-User Access:** Any modern web browser (Google Chrome, Microsoft Edge, Safari) or the Power BI Mobile App (iOS/Android) for viewing the report.

CHAPTER 4

METHODOLOGY

The methodology of the "Netflix Data Analysis Dashboard using Power BI" focuses on transforming raw Netflix data into meaningful and interactive visual insights. The process begins with data collection, where the dataset is obtained from Kaggle in CSV format. This dataset contains information such as title, type, genre, country, rating, and release year, which forms the foundation for analysis.

The next step involves data preparation using Power Query. In this stage, the raw data is cleaned by removing null values, correcting inconsistencies, and converting columns into appropriate data types. Additional transformations such as extracting year from date fields and organizing categorical data are performed to ensure the dataset is structured properly for analysis.

4.1 Data Collection

The data used in this project is collected from Kaggle, a popular platform for publicly available datasets. The dataset is titled "Netflix Movies and TV Shows" and contains real-world information about Netflix content. It is provided in CSV format, which makes it easy to import into Power BI for analysis and visualization. The dataset includes important attributes such as show ID, title, type (Movie or TV Show), genre, country, rating, and release year, which are essential for understanding content distribution and trends.

4.2 Data Preparation (Power Query)

Power Query was utilized to clean and transform the raw Netflix dataset to ensure accuracy and performance. The key steps included:

● Handling Null Values: Removed or replaced missing data in columns such as country, rating, and to avoid errors during analysis.

● Standardizing Columns: Renamed column headers into simple and user-friendly names for better understanding and easier dashboard navigation.

● Formatting Data Types: Converted fields like release_year into numeric format and ensured categorical fields such as genre, type, and rating were properly classified.

4.3 Data Visualization

Key Performance Indicators (KPI Cards) → Display high-level metrics at a glance, including Total Titles, Total Genres, Total Ratings, and Year Range of content.

Line Chart → Illustrates the trend of Netflix content growth over the years, showing how the number of movies and TV shows has increased over time.

Bar Charts → Represent the distribution of content across different genres and ratings, helping to identify the most popular categories.

Treemap → Displays country-wise content distribution, highlighting which countries contribute the most to Netflix content.

Donut/Pie Chart → Shows the proportion of Movies vs TV Shows, making it easy to compare the content types on Netflix.

4.4 Filters and Interactivity

To make the dashboard interactive, slicers are added:

- Year
- Country
- Genre
- Type (Movie / TV Show)

These filters allow users to dynamically explore the dataset and perform customized analysis based on selected parameters.

4.5 Dashboard Design

The final dashboard is designed to present all insights in a structured layout:

Top Section: Displays Key Performance Indicators (KPIs) such as Total Titles, Total Genres, Total Ratings, and Year Range for quick insights.

Left Panel: Contains interactive filters/slicers to refine data by country, genre, year, and content type (Movie/TV Show).

Center Area: Shows main visualizations such as bar charts and treemaps to analyze genre distribution and country-wise content contribution.

Right & Bottom Sections: Include comparison and trend visuals like donut charts (Movies vs TV Shows) and line charts to analyze content growth over the years.

The structured layout ensures easy navigation and clear understanding of Netflix content distribution and trends.

4.6 Outcome

Interactive Dashboard: Development of an interactive Power BI dashboard for easy analysis of Netflix movies and TV shows data.

● Content Distribution Insights: Clear visualization of content based on genres, ratings, and countries.

● Trend Analysis: Effective tracking of content growth over the years, showing how Netflix content has expanded over time.

● Comparative Analysis: Ability to compare Movies vs TV Shows and analyze distribution across different genres and regions.

● Efficient Data Processing: Implementation of a structured ETL process using Power Query and DAX for clean and reliable data.

● Decision Support: Provides meaningful insights to understand content patterns and supports data-driven analysis.

CHAPTER 5
IMPLEMENTATION

The implementation of the Netflix Data Analysis Dashboard involves converting raw dataset into meaningful insights using Power BI. Initially, the dataset is imported into Power BI using the "Get Data" option and verified for accuracy. Data transformation is performed in Power Query, where missing values are handled, columns are cleaned, and appropriate data types are assigned. A simple data model is used to organize the dataset efficiently. DAX measures are then created to calculate key metrics such as total titles, genre count, and rating distribution. Various visualizations like KPI cards, bar charts, treemaps, donut charts, and line charts are developed to represent different aspects of the datax. Finally, the dashboard is designed with a structured layout and consistent formatting to ensure clarity, usability, and effective presentation of insights.

5.1 Dataset Description

The dataset used in this project is the "Netflix Movies and TV Shows" dataset obtained from Kaggle. It contains structured information about the content available on Netflix, including both movies and TV shows. The dataset is provided in CSV format and serves as the primary data source for analysis and visualization in Power BI.

The dataset includes several important attributes such as show_id (unique identifier), title (name of the content), type (Movie or TV Show), country (production country), listed_in (genre/category), rating (content classification like TV-MA, PG, etc.), and release_year (year of release). These fields are used to analyze content distribution, identify popular genres and ratings, study trends over time, and compare different categories

5.2 Data Loading

- The dataset is loaded into Power BI using the **"Get Data"** option.
- Data is previewed and verified before processing.

5.3 Data Transformation (Power Query)

Using Power Query Editor, the dataset is cleaned and prepared:

Removed missing and null values from important columns such as country, rating, and genre to ensure accurate analysis.

● Renamed and standardized column names to make them simple and user-friendly for better understanding.

● Converted data types by setting release_year as numeric and ensuring categorical fields like type and rating are properly formatted.

● Organized and structured the dataset to enable efficient filtering and smooth interaction in the dashboard.

5.4 Data Modeling

- **Model Structure:** The project uses a **single-table (flat schema)** where all geographical, temporal, and energy-related data is stored in one central table for simplicity and efficiency.
- **Relationships:** Since the data resides in a single table, Power BI internally manages relationships between fields like Continent, Area, and Year without requiring separate tables.
- **Drill-Down Hierarchy:** A geographical hierarchy (Continent → Area) is implemented to allow users to move from global insights to country-level analysis easily.
- **Efficiency:** This simplified model improves performance, enables smooth cross-filtering, and reduces complexity in dashboard design

5.5 DAX Measures (Key Calculations)

To generate insights, the following measures are created:

```
Total Titles = COUNT('Netflix'[show_id])

● Total Genres = DISTINCTCOUNT('Netflix'[listed_in])

● Total Ratings = DISTINCTCOUNT('Netflix'[rating])

● Movies Count = CALCULATE(COUNT('Netflix'[show_id]), 'Netflix'[type] = "Movie")

● TV Shows Count = CALCULATE(COUNT('Netflix'[show_id]), 'Netflix'[type] = "TV Show")

These measures are used in charts and KPI visuals to display key insights

about Netflix content.
```

5.6 Dashboard Visualizations

Main Visuals

The dashboard includes various visual elements to represent Netflix data effectively:

● KPI Cards: Display key metrics such as Total Titles, Total Genres, and Total Ratings for quick insights.

● Bar Charts: Show the distribution of content across different genres and ratings, helping to identify the most popular categories.

● Treemap: Represents country-wise content contribution, highlighting which countries produce the most Netflix content.

● Donut Chart: Compares Movies and TV Shows, showing the proportion of each content type.

● Line Chart: Illustrates the trend of content growth over the years, helping to understand how Netflix content has increased over time.

● Interactive Slicers: Enable filtering based on year, country, genre, and type to provide dynamic and customized analysis.

5.7 Filters / Slicers

To enhance interactivity, the following filters are added:

● Country: Enables selection of specific countries for detailed analysis of content distribution.

● Genre: Allows users to filter and analyze content based on different genres such as Drama, Comedy, etc.

● Type (Movie/TV Show): Provides filtering to compare Movies and TV Shows.

● Year: Supports time-based filtering to analyze content trends over different years.

● Rating: Lets users filter content based on classification such as TV-MA, PG, etc.

These slicers enable dynamic filtering, helping users explore Netflix data, compare categories, and analyze trends effectively.

5.8 Dashboard Layout

The dashboard is organized in a structured manner to ensure seamless navigation and high readability:

● Top → KPI Cards (Total Titles, Total Genres, Total Ratings, and Year Range).

● Left side → Filters (Country, Genre, Type, and Year slicers).

● Center → Main Visualizations (Bar charts and treemap showing genre distribution and country-wise content).

● Right & Bottom → Comparison & Trends (Donut chart for Movies vs TV Shows and line chart for yearly content growth).

This layout ensures a logical flow, allowing users to move from high-level summaries to detailed insights about Netflix content.

5.9 Integration

- Data is imported from Excel/CSV
- All visuals are connected to the same dataset
- Filters dynamically update all visuals

5.10 Final Output

The final output of this project is an interactive and visually appealing Power BI dashboard that provides comprehensive insights into Netflix movies and TV shows data. The dashboard effectively transforms raw data into meaningful visual representations using KPI cards, bar charts, treemaps, donut charts, and line charts.

The dashboard allows users to analyze content distribution based on genres, ratings, countries, and release years. With the help of interactive filters and slicers, users can dynamically explore the dataset and gain customized insights. It highlights key patterns such as the dominance of movies over TV shows, popular genres, and the growth of content over time.

Overall, the final output is a user-friendly dashboard that simplifies complex data and supports better understanding and analysis of Netflix content trends.

CHAPTER 6

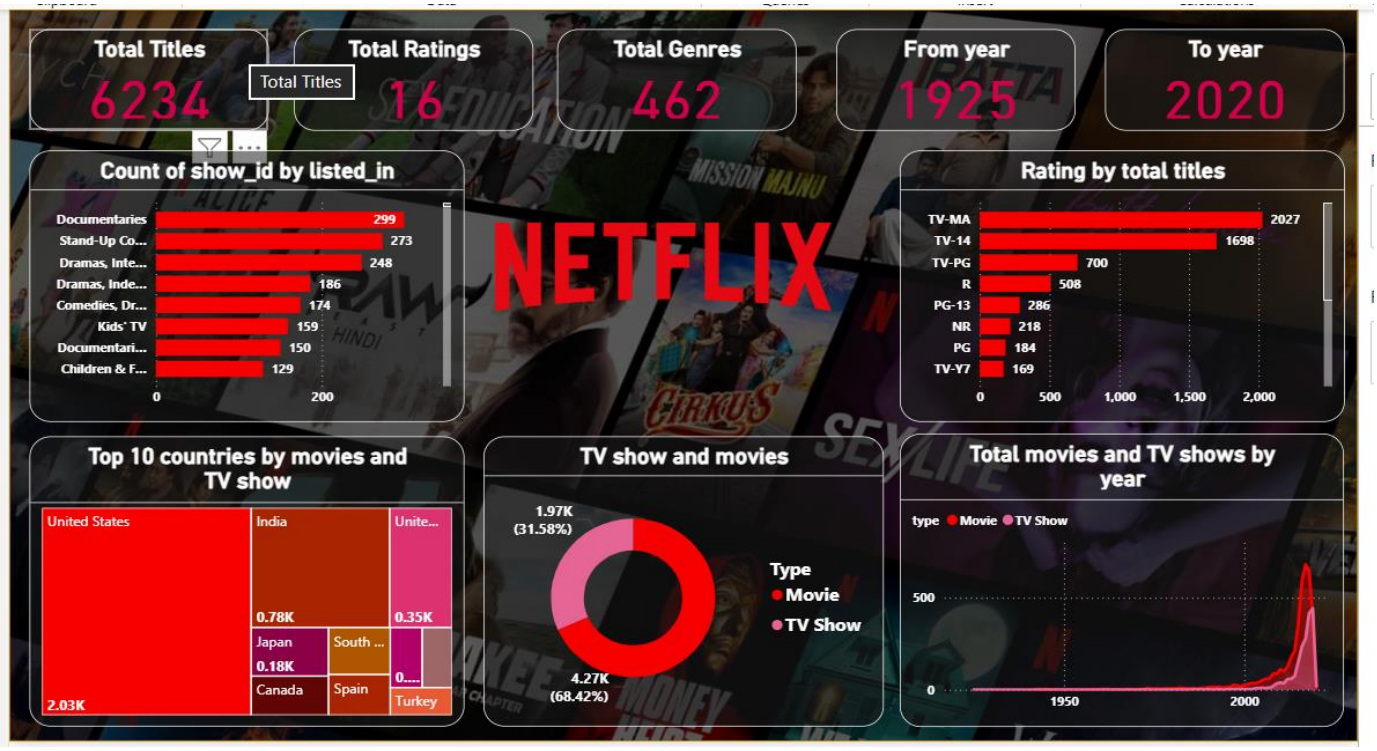
RESULT

The developed “Netflix Data Analysis Dashboard using Power BI” successfully provides an interactive and visually rich representation of Netflix content data. The dashboard effectively transforms raw dataset into meaningful insights using various visualization techniques such as KPI cards, bar charts, treemaps, donut charts, and line charts. These visuals help in understanding patterns, trends, and distributions within the dataset in a clear and simplified manner.

The KPI cards display key metrics such as total number of titles, total genres, and total ratings, providing a quick overview of the dataset. The bar charts highlight the distribution of content across different genres and ratings, making it easy to identify the most popular categories. The treemap visualization clearly shows the country-wise contribution of Netflix content, indicating that certain countries dominate in content production. The donut chart compares Movies and TV Shows, revealing that movies constitute a larger portion of the content compared to TV shows.

The line chart illustrates the growth of Netflix content over the years, showing a significant increase in the number of releases after the year 2000. This indicates the rapid expansion of Netflix as a streaming platform and the growing demand for digital content. The dashboard also includes interactive filters and slicers such as country, genre, type, and year, allowing users to dynamically explore the dataset and perform customized analysis.

Overall, the project achieves its objective of providing a clear, interactive, and insightful analysis of Netflix data. The dashboard enables users to identify trends, compare different categories, and gain valuable insights into content distribution and growth. It demonstrates the effectiveness of Power BI as a tool for data visualization and highlights the importance of converting raw data into meaningful information for better decision-making.


Total Titles
6234
Total Ratings
16
Total Genres
462
From year
1925
To year
2020
Total Titles
Count of show_id by listed_in
Documentaries 299
Stand-Up Co... 273
Dramas, Inte... 248
Dramas, Inde... 186
Comedies, Dr... 174
Kids' TV 159
Documentari... 150
Children & F... 129
0
200
NETFLIX
Rating by total titles
TV-MA 2027
TV-14 1698
TV-PG 700
R 508
PG-13 286
NR 218
PG 184
TV-Y7 169
0
500
1,000
1,500
2,000
Top 10 countries by movies and TV show
United States
2.03K
India
0.78K
Unite...
0.35K
Japan
0.18K
South ...
Canada
Spain
Turkey
TV show and movies
1.97K
(31.58%)
4.27K
(68.42%)
Type
Movie
TV Show
Total movies and TV shows by year
type Movie TV Show
500
0
1950
2000

CONCLUSION

The “Netflix Data Analysis Dashboard using Power BI” project successfully demonstrates how raw data can be transformed into meaningful insights through effective data visualization techniques. By utilizing Power BI, Power Query, and DAX, the project converts a complex Netflix dataset into an interactive and user-friendly dashboard that provides clear understanding of content distribution, trends, and patterns.

The dashboard highlights key aspects such as the dominance of movies over TV shows, the popularity of certain genres like Drama and Comedy, and the significant contribution of countries such as the United States in content production. It also reveals the steady growth of Netflix content over the years, particularly after 2000, indicating the rapid expansion of the platform and increasing demand for digital streaming services.

Through the use of interactive filters and slicers, users can explore the data dynamically based on parameters like country, genre, type, and year. This enhances the usability of the dashboard and allows for detailed and customized analysis. The visualizations make it easy to interpret large amounts of data quickly and effectively, supporting better understanding and decision-making.

Overall, this project emphasizes the importance of data visualization in analyzing real-world datasets. It showcases how tools like Power BI can simplify complex information and present it in an organized and visually appealing manner. The Netflix dashboard serves as a valuable analytical tool and provides a strong foundation for further enhancements such as real-time data integration and predictive analytics.

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